Concentration

Markov $X \geq 0$, $P(X \geq t) \leq \frac{1}{t} E(X)$

Chebyshev $P(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}$

Sub-Gaussian : Gaussian、±1 Bernoulli

Hoeffding for Bernoulli $P(a^T X \geq t) \leq e^{-\frac{t^2}{2\|a\|^2}}$

$P(|a^T X| \geq t) \leq 2e^{-\frac{t^2}{2\|a\|^2}}$

Sub-Exp : Exponential . χ^2

Berstein's $P(|a^T X| \geq t) \leq 2e^{-CN\min(\frac{t^2}{K^2}, \frac{t}{K})}$ $K = \max \|X_i\|_{\psi_1}$

Hanson-Wright

$P(|X^T A X| \geq t) \leq 2e^{-\frac{1}{\|A\|_F^2} C \min(\frac{t^2}{K^2}, \frac{t}{K})}$ $K = \max \|X_i\|_{\psi_2}$

χ^2 分布 $X = X_1^2 + \cdots + X_n^2$

$f(x) = \frac{1}{2^{n/2}\Gamma(\frac{n}{2})} x^{\frac{n}{2}-1} e^{-\frac{1}{2}x}$ $E(X) = n$ $D(X) = 2n$

Chi 分布 $X = \sqrt{X_1^2 + \cdots + X_n^2}$

$f(x) = \frac{1}{2^{n/2-1}\Gamma(\frac{n}{2})} x^{n-1} e^{-\frac{1}{2}x}$ $E(X) = \sqrt{2}\frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2})}$

Gaussian 绝对值 $X = |X_1|$: $n=1$ 的 Chi 分布

$E(X) = \sqrt{2}/\Gamma(\frac{1}{2})$

Rayleigh 分布 $X = \sqrt{X_1^2 + X_2^2}$: $n=2$ 的 Chi 分布

$f(x) = \frac{1}{\sigma^2} x^2 e^{-\frac{x^2}{2\sigma^2}}$ $E(X) = \sqrt{\pi/2}\,\sigma^2$ $D(X) = \frac{4-\pi}{2}\sigma^2$

复 Gaussian $X = X_1 + jX_2$

$f(x) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\{-\frac{x_1^2}{2\sigma_1^2} - \frac{x_2^2}{2\sigma_2^2}\}$ $D(X) = \begin{bmatrix} \sigma_1^2 & \\ & \sigma_2^2 \end{bmatrix}$ $P = E(|X|^2) = \sigma_1^2 + \sigma_2^2$

Concentration

Markov $X \geq 0$, $P(X \geq t) \leq \frac{1}{t} E(X)$

Chebyshev $P(|X-\mu| \geq t) \leq \frac{\sigma^2}{t^2}$

Pf : $P[(X-\mu)^2 \geq t^2] \leq \frac{1}{t^2} E[(X-\mu)^2]$

Sub-Gaussian

Def: $P(|X|>t) \le 2e^{-C^2t^2}$

Gaussian、±1 Bernoulli

Pf: $P(|X|>t) \le 2\frac{1}{\sqrt{2\pi}\,t}e^{-\frac{t^2}{2\sigma^2}}$

$$P(X>t) = \frac{1}{\sqrt{2\pi}\,\sigma}\int_t^\infty e^{-\frac{x^2}{2\sigma^2}}dx \le \frac{1}{\sqrt{2\pi}\,\sigma}\int_t^\infty \frac{x}{t}e^{-\frac{x^2}{2\sigma^2}}dx = \frac{1}{\sqrt{2\pi}\,t}e^{-\frac{t^2}{2\sigma^2}}$$

Hoeffding for ±1 Bernoulli

$P(|X_1+\cdots+X_N| \ge t) \le 2e^{-\frac{t^2}{2N}}$ $\qquad$ $P(|a^TX| \ge t) \le 2e^{-\frac{t^2}{2\|a\|^2}}$

[例] $N=2$, $X = X_1 + X_2$

X	-2	0	2
P	1/4	1/2	1/4

$P(|X| \ge 1) = \frac{1}{2} \le 2e^{-1}$

Hoeffding for 0/1 Bernoulli

$P(|X_1+\cdots+X_N - \frac{N}{2}| \ge t) \le 2e^{-\frac{2t^2}{N}}$

Hoeffding for Gaussian

ψ_2-Norm $\quad \|X\|_{\psi_2} = \inf\{t \ge 0,\ E(e^{\frac{X^2}{t^2}}) \le 2\}$

$P(|X_1+\cdots+X_N| \ge t) \le 2e^{-Ct^2/\sum\|X_i\|_{\psi_2}^2}$

Sub-Exp

Def: $P(|X|>t) \le 2e^{-Ct}$ $\qquad$ Exponential. χ^2

Berstein's

$P(|X_1+\cdots+X_N| \ge t) \le 2e^{\frac{\frac{1}{2}t^2}{\sigma^2+\frac{1}{3}Kt}}$ $\qquad$ 0-mean

$P(|a^TX| \ge t) \le 2e^{-CN\min(\frac{t^2}{K^2},\frac{t}{K})}$ $\qquad$ $K = \max\|X_i\|_{\psi_1}$

Hanson-Wright

$P(|X^TAX - E[X^TAX]| \ge t) \le 2e^{-\frac{1}{\|A\|_F^2}C\min(\frac{t^2}{K^2},\frac{t}{K})}$ $\qquad$ $K = \max\|X_i\|_{\psi_2}$